

Tight Tangent Parabolas Characterize 3-Convexity

A full solution of Problem 1 in arXiv:2305.04353

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Model: GPT5.6

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Status: full solution, likely valid. The answer to Problem 1 is affirmative under exactly the stated continuity and differentiability assumptions. Expert review should focus on the local BV/mollification bridge in Lemma 2.

1 Source question

Marinescu and Niculescu ask the following in Problem 1 on page 9 of [1]. A *tight tangent parabola* at $c \in \text{int } I$ is a quadratic polynomial P_c tangent to f at c such that

$$P_c(x) \geq f(x) \quad (x \leq c), \quad P_c(x) \leq f(x) \quad (x \geq c),$$

whenever the points lie in I .

Suppose that f is continuous on I and differentiable on $\text{int } I$, and admits at every point $c \in \text{int } I$ a tight tangent parabola. Is f necessarily 3-convex?

Here 3-convexity means nonnegativity of every third divided difference $[x_0, x_1, x_2, x_3; f]$ for $x_0 < x_1 < x_2 < x_3$ in I .

As a consequence,

$$f(x) = \sup \left\{ f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2}y : a \in \text{int } I, a < x, y \in \partial f'(a) \right\}$$

for all $x \in I$ different from the left endpoint of I .

Geometrically, $g = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2}f''_+(a)$ (respectively $y = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2}f''_-(a)$) represents the unique parabola tangent to the graph of f at a and which has the same right-hand (left-hand) derivative of second order at a . These "tight tangent parabolas" represent analogs of the line supports from the theory of convex functions. Notice that a tight tangent parabola at a point a is above the graph of f on $I \cap (-\infty, a]$ and under the graph on $I \cap [a, \infty)$.

Problem 1. Suppose that f is a function continuous on I and differentiable on $\text{int } I$, which admit at every point $a \in \text{int } I$ a tight tangent parabola. Is f necessarily 3-convex?

Figure 1: The tightness convention and complete Problem 1, rendered from page 9 of the source PDF.

2 Proof intuition

At symmetric points around c , the even part of any tangent parabola cancels. Its one-sided position relative to the graph therefore forces the central secant slope of f to dominate $f'(c)$. For a C^1 function this is close to a standard characterization of 3-convexity, but the problem assumes only differentiability and a derivative need not be continuous.

The regularity gap can nevertheless be closed without strengthening the hypotheses. A fixed-radius central-secant bound makes f' locally bounded above. Hence f is locally of bounded variation and the singular part of its distributional derivative is nonpositive. Convolution with a nonnegative mollifier can only lower the derivative, so the central-secant inequality survives smoothing. For the smooth approximants it immediately forces a nonnegative third derivative. Passing to the limit completes the argument.

3 The affirmative solution

Lemma 1 (The midpoint inequality). *Under the hypotheses of Problem 1, for every $c \in \text{int } I$ and every $h > 0$ for which $c - h, c + h \in I$,*

$$f'(c) \leq \frac{f(c+h) - f(c-h)}{2h}. \quad (1)$$

Proof. Write the tight tangent parabola at c as

$$P_c(x) = f(c) + f'(c)(x-c) + A_c(x-c)^2.$$

Tightness gives $P_c(c-h) \geq f(c-h)$ and $P_c(c+h) \leq f(c+h)$. Subtracting these two inequalities yields

$$f(c+h) - f(c-h) \geq P_c(c+h) - P_c(c-h) = 2hf'(c),$$

because the constant and quadratic terms cancel. □

Lemma 2 (Regularization preserves the midpoint inequality). *Let U be an open interval and let $f : U \rightarrow \mathbb{R}$ be differentiable and satisfy (1) whenever the symmetric interval is contained in U . On every compact subinterval of U , the standard nonnegative mollifications f_ε are defined for all sufficiently small ε and satisfy the same inequality. Consequently,*

$$f_\varepsilon''' \geq 0.$$

Proof. Work locally. Choose compact intervals $J \subset \text{int } K \subset U$. There is an $h_0 > 0$ such that $x \pm h_0 \in U$ for every $x \in K$. Applying (1) with this fixed h_0 , and using continuity of f on the relevant compact set, gives a constant M such that $f'(x) \leq M$ on K .

The mean value theorem now shows that $x \mapsto f(x) - Mx$ is nonincreasing on K . Thus f is of bounded variation there. In the local Lebesgue decomposition of its distributional derivative,

$$Df = f' \mathcal{L}^1 + \nu, \quad (2)$$

the singular measure ν is nonpositive. Indeed, the absolutely continuous density is the classical derivative almost everywhere, while the singular part of $D(f - Mx) \leq 0$ is precisely ν .

Let $\rho_\varepsilon \geq 0$ be a smooth standard mollifier, with support small enough that all convolutions below are local to K , and put $f_\varepsilon = f * \rho_\varepsilon$ on J . From (2),

$$f'_\varepsilon = (f' \mathcal{L}^1) * \rho_\varepsilon + \nu * \rho_\varepsilon \leq f' * \rho_\varepsilon. \quad (3)$$

For every sufficiently small $h > 0$, convolving (1) and using commutation with translations gives

$$f' * \rho_\varepsilon(x) \leq \frac{f_\varepsilon(x+h) - f_\varepsilon(x-h)}{2h}.$$

Together with (3), this is (1) for f_ε . Taylor expansion at x gives

$$\frac{f_\varepsilon(x+h) - f_\varepsilon(x-h)}{2h} = f'_\varepsilon(x) + \frac{h^2}{6} f_\varepsilon'''(x) + o(h^2).$$

Subtract $f'_\varepsilon(x)$, divide by h^2 , and let $h \downarrow 0$ to obtain $f_\varepsilon'''(x) \geq 0$. □

Theorem 1. *Let I be an interval and let $f : I \rightarrow \mathbb{R}$ be continuous on I and differentiable on $\text{int } I$. If f admits a tight tangent parabola at every point of $\text{int } I$, then f is 3-convex on I .*

Proof. Lemma 1 gives (1). Fix four points $x_0 < x_1 < x_2 < x_3$ in $\text{int } I$ and choose compact $J \subset \text{int } I$ containing them in its interior. By Lemma 2, each sufficiently fine mollification has $f_\varepsilon''' \geq 0$ near J . The generalized mean value theorem for divided differences therefore gives

$$[x_0, x_1, x_2, x_3; f_\varepsilon] \geq 0.$$

As $\varepsilon \downarrow 0$, continuity of f gives locally uniform convergence $f_\varepsilon \rightarrow f$. A divided difference at fixed distinct nodes is a finite linear combination of the four function values, hence

$$[x_0, x_1, x_2, x_3; f] \geq 0.$$

This proves 3-convexity for interior nodes. If I contains one or both endpoints, approximate them by interior nodes and use continuity once more. $\square$

4 What the proof uses—and does not use

The coefficient A_c may depend arbitrarily on c and need not be locally bounded. It disappears in Lemma 1. No second derivative, absolute continuity, or a priori continuity of f' is assumed. Possible singular BV behavior is retained in the measure ν ; its sign, rather than its absence, is what makes the smoothing step work.

5 Verification and novelty status

The formal verification report checks the cancellation, the BV decomposition, the convolution inequality, the Taylor limit, and passage of divided differences to the limit. The principal human-review point is the sign of the singular part in (2); it follows locally from $D(f - Mx) \leq 0$.

Before promotion, the run indexes were searched by arXiv id, title, and core terms. Bounded web searches on 26 August 2026 used the exact title, the arXiv id, the quoted problem, both authors, and combinations of “tight tangent parabola” with “3-convex”. They found the source and its journal version, but no distinct later paper explicitly resolving Problem 1. This is bounded novelty evidence, not a claim of exhaustive priority.

6 Human-review recommendation

The packet should be reviewed as a claimed full affirmative solution. The midpoint cancellation is elementary; the only nontrivial audit item is the local BV regularization in Lemma 2. In particular, a review should confirm both inequality directions in (3) before publication or external circulation.

References

- [1] D.-S. Marinescu and C. P. Niculescu, *Old and new on the 3-convex functions*, arXiv:2305.04353, 2023.
- [2] L. C. Evans and R. F. Gariepy, *Measure Theory and Fine Properties of Functions*, revised ed., CRC Press, 2015. Standard reference for BV derivatives, Lebesgue decomposition, and mollification.